

Section A

For each question there are four possible answers, **A**, **B**, **C** or **D**. Choose the **one** you consider to be correct.

- 1 The relative mass of unknown element **R** consists of five isotopes. The table shows the relative abundance of these five isotopes.

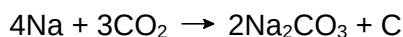
Isotope	Relative abundance (%)
^{90}R	50
^{91}R	10
^{92}R	15
^{94}R	20
^{96}R	5

What is the relative atomic mass of **R**?

- A** 90.00 **B** 91.50 **C** 92.60 **D** 93.00

- 2 *Use of the Data Booklet is relevant to this question.*

Sodium reacts with carbon dioxide to produce sodium carbonate and carbon only.



If all the $1.1 \times 10^7 \text{ dm}^3$ carbon dioxide, measured at standard temperature and pressure, could be reacted with sodium, what would be the mass of sodium carbonate produced?

- A** $3.2 \times 10^7 \text{ g}$ **B** $3.5 \times 10^7 \text{ g}$
C $7.3 \times 10^7 \text{ g}$ **D** $7.8 \times 10^7 \text{ g}$

- 3 10 cm^3 of propane is burned in 70 cm^3 of oxygen (an excess). What is the maximum theoretical decrease in volume of the gases? All gaseous volumes were measured at room temperature and pressure.

- A** 80 cm^3 **B** 50 cm^3 **C** 30 cm^3 **D** 10 cm^3

- 4 25 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ $\text{X}(\text{NO}_3)_2$ solution required 10 cm^3 of $0.100 \text{ mol dm}^{-3}$ of acidified potassium manganate(VII) solution for complete reaction. What is the change in oxidation state of element **X**?

- A** +2 **B** +3 **C** +4 **D** +5

- 5 The successive ionisation energies (IE) of two elements, **Y** and **Z**, are given below.

IE / kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Y	1090	2350	4610	6220	37800	47000	-	-
Z	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **Y** reacts with **Z**?

- A** YZ **B** Y₂Z₃ **C** YZ₄ **D** Y₄Z

- 6 Which of the following species are the number of electrons and neutrons equal?

- A** ${}^9_4\text{Be}$ **B** ${}^{35}_{17}\text{Cl}^+$ **C** ${}^{18}_8\text{O}^{2-}$ **D** ${}^{24}_{12}\text{Mg}^{2+}$

- 7 A solid **M** has the following properties.

- It melts at 2850°C and boils at 3500°C.
- It conducts electricity in molten state but not in solid state.

What could be the likely structure of solid **M**?

- A** an ionic crystal lattice **B** an atomic crystal lattice
C a molecular crystal lattice **D** a giant molecular crystal lattice

- 8 Which one of the following pairs of compounds contains two species with different shapes?

- A** PCl₃ and BrCl₃ **B** BeCl₂ and BrCl₂⁻
C CO₃²⁻ and BF₃ **D** SiF₄ and PF₄⁺

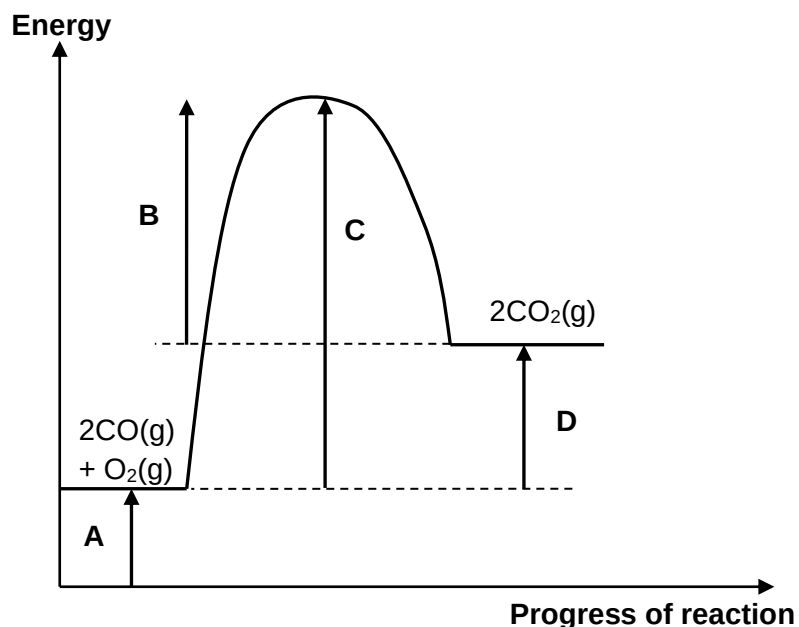
- 9 Hydrogen bonding can occur between molecules of methanal, HCHO, and molecules of liquid **T**.

What could liquid **T** be?

- A** CH₃OH **B** CH₃CHO
C CH₃COCH₃ **D** CH₃CO₂CH₃

- 10 The reaction pathway diagram for the reaction $2\text{CO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)}$ is shown below.

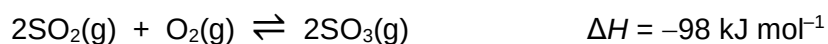
Which of the following arrows correctly represents the enthalpy change of reaction?



- 11 In a calorimetric experiment, 1.60 g of ethane is burnt. 45% of the energy released is absorbed by 200 g of water whose temperature rose from 18 °C to 66 °C. The specific heat capacity of water is 4.20 J g⁻¹ K⁻¹.

What is the enthalpy change of combustion of ethane?

- A $-4.72 \times 10^5 \text{ J mol}^{-1}$ B $-1.05 \times 10^6 \text{ J mol}^{-1}$
 C $-1.68 \times 10^6 \text{ J mol}^{-1}$ D $-2.70 \times 10^6 \text{ J mol}^{-1}$
- 12 Which one of these statements can be deduced from the following information?



- A Increasing the pressure increases the value of the equilibrium constant.
 B Increasing the temperature decreases the rate of the forward reaction.
 C Increasing the pressure decreases the proportion of SO₂ and O₂ at equilibrium.
 D Adding a catalyst increases the yield and rate of production of SO₃.

- 13** Hydrogen peroxide, H_2O_2 decomposes when exposed to light into oxygen gas and water according to the rate equation below:
Rate = $k[\text{H}_2\text{O}_2]$

1.0 mole of H_2O_2 was allowed to decompose in a bottle. At the end of 10 hours, it was found that only 25% of the original sample was left in the reaction flask. What is the half-life of the decomposition of H_2O_2 ?

- | | | | |
|----------|-----------|----------|------------|
| A | 2.5 hours | B | 5.0 hours |
| C | 7.5 hours | D | 10.0 hours |

- 14** NO_2 dimerises to form N_2O_4 as follows:



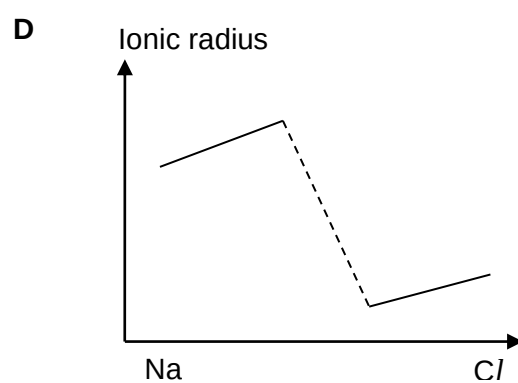
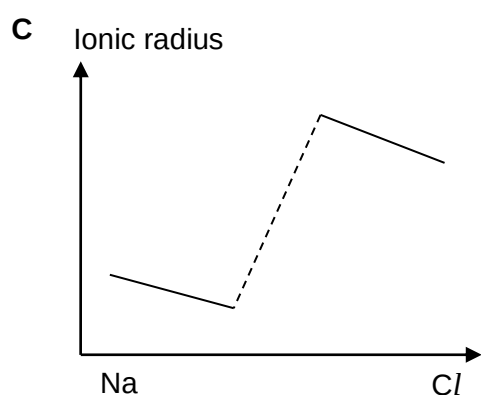
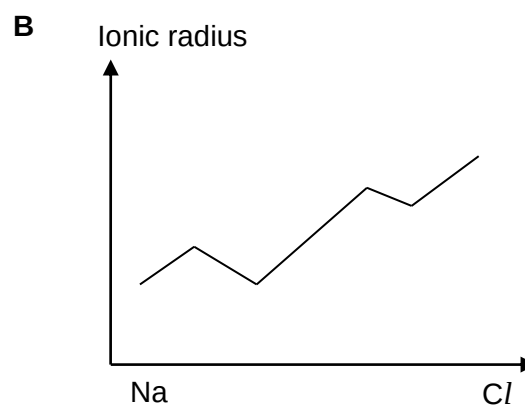
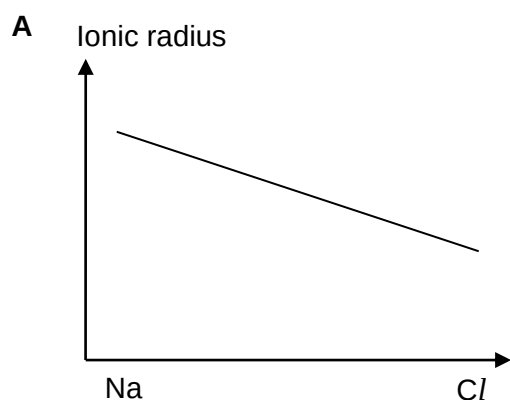
Which of the following change would affect the equilibrium position of the above reaction?

- A** addition of catalyst
- B** cooling the mixture
- C** an increase in temperature
- D** an increase in pressure

- 15** Which equation represents the reaction of sulfur dioxide with an excess of aqueous sodium hydroxide?

- A** $\text{SO}_2 + \text{NaOH} \rightarrow \text{NaHSO}_3$
- B** $\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O} + \text{O}_2$
- C** $\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$
- D** $\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2$

- 16** Which diagram represents the variation of ionic radius for the ions of the elements across the third period (Na to Cl)?

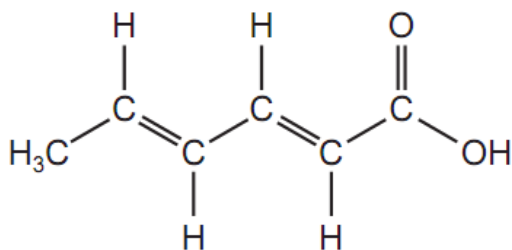


- 17** The following elements are in the third period of the Periodic Table.

Which has the highest melting point?

- A** aluminium **B** magnesium **C** sodium **D** sulfur

- 18 Sorbic acid is used as a food preservative because it kills fungi and moulds.

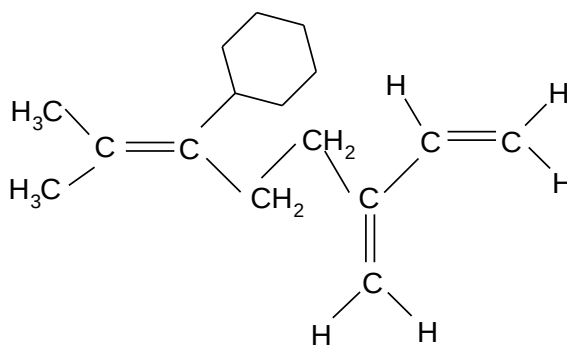


sorbic acid

Identify the number of sp^2 and sp^3 hybridised carbon in sorbic acid.

	sp^2	sp^3
A	2	4
B	4	2
C	1	5
D	5	1

- 19 How many pairs of geometric isomers do the following unsaturated hydrocarbon has?

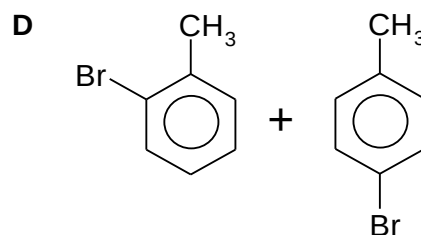
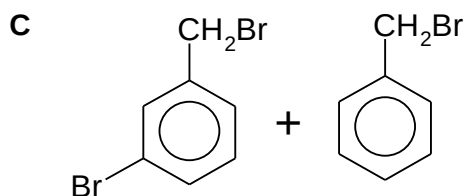
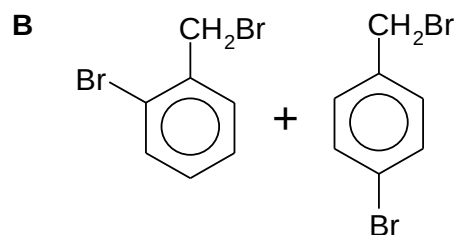
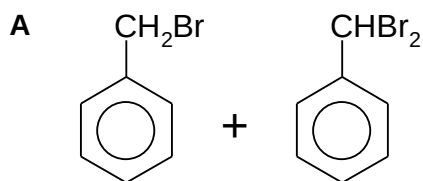


- A 0
C 4

- B 2
D 6

- 20** Methylbenzene reacts with bromine in the presence of a suitable halogen carrier in the dark.

Identify the main products of the reaction if equimolar quantities of methylbenzene reacts with bromine.



- 21** Chlorofluorocarbons, CFCs are widely used but can deplete the ozone layer.

A typical CFC is CHF_2CHClF . Which covalent bond in this CFC is the weakest?

- A** C-C
- B** C-F
- C** C-H
- D** C-Cl

- 22** For which pair of compounds can alkaline aqueous iodine be used to distinguish between members of the pair?

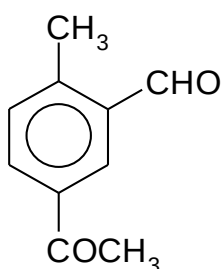
- A** $\text{CH}_3\text{CO}_2\text{H}$ and CH_3COCH_3
- B** CH_3CHO and CH_3COCH_3
- C** $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
- D** $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- 23** A sample of propanal is treated with HCN in the presence of a little NaCN. The organic product is then heated under reflux with dilute sulfuric acid.

What will be the final organic product?

- A** $\text{CH}_3\text{CH}_2=\text{CHCO}_2\text{H}$
B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CO}_2\text{H}$
C $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CN}$
D $\text{CH}_3\text{CH}_2\text{C}(\text{OH})(\text{CN})\text{CH}_3$

- 24** The following compound is tested with Fehling's and Tollen's reagent separately.



What are the expected observations?

warm with Fehling's reagent	warm with Tollen's reagent
brick-red precipitate	silver mirror
no precipitate	silver mirror
brick-red precipitate	no silver mirror
no precipitate	no silver mirror

- 25 Which of the following compounds **cannot** be used as a starting material to synthesise carboxylic acid in a single step?

- A** $\text{CH}_3\text{CH}_2\text{OH}$ **B** $\text{C}_6\text{H}_5\text{Br}$
C $\text{CH}_3\text{CH}_2\text{CHO}$ **D** CH_3CN

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

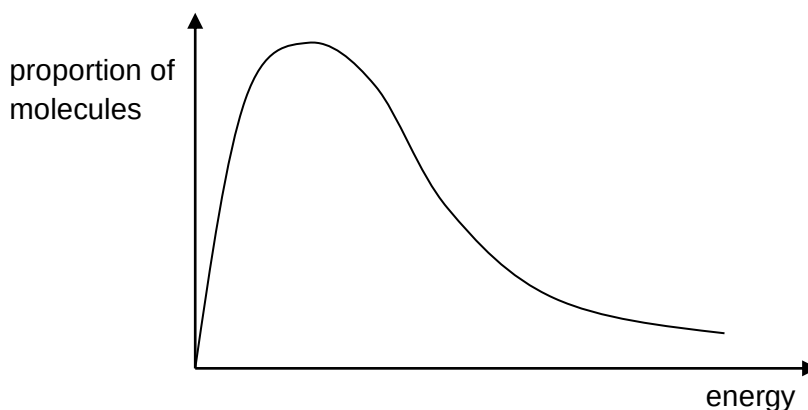
No other combination of statements is used as a correct response.

- 26** Boron forms a compound with nitrogen known as boron nitride, BN which has a structure similar to graphite.

Which conclusions can be drawn from this information?

- 1** Boron nitride is a good lubricant.
- 2** The boron and nitrogen atoms are likely to be arranged in a hexagonal pattern.
- 3** Boron nitride has a layer structure with van der Waals' forces between layers.

- 27** The diagram represents the Boltzmann distribution of molecular energies at a given temperature.



As temperature increases, which statements are correct?

- 1** The maximum of the curve is displaced to the right.
- 2** The proportion of molecules with energies **above** any given value increases.
- 3** The proportion of molecules **with** any given energy increases.

The responses **A** to **D** should be selected on the basis

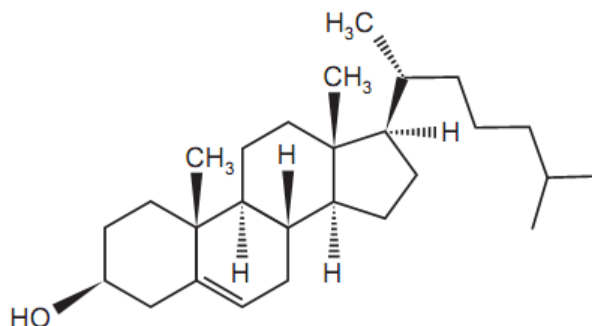
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 Which of the following chlorides would produce hazardous gases in case of spillage into a reservoir?

- 1** silicon chloride
- 2** magnesium chloride
- 3** sodium chloride

29 The diagram shows the structure of the naturally-occurring molecule cholesterol.



cholesterol

Which statements about cholesterol are correct?

- 1** The molecule contains a secondary alcohol.
- 2** The molecule reacts with 2,4-dinitrophenylhydrazine.
- 3** The molecule is able to exhibit geometric isomerism.

30 Which starting materials will give sodium benzoate on heating with aqueous sodium hydroxide under reflux?

- 1** methylbenzene and potassium dichromate
- 2** phenyl benzoate
- 3** phenyl nitrile

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